

Werner Bisschoff

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Software Engineer

Professional Summary

Software engineer with 5+ years of experience bridging hardware and full-stack development. I combine technical depth in C++ and Python with a rigorous, agentic engineering methodology. My focus is on resilient system design and productivity-enhancing AI workflows, governed by security, processes, and specification-driven development.

Experience

Founder and Host , Cape Town

Divergent Tabletop | Jul 2025

- Founded neurodivergent-focused peer community; manage event operations and WhatsApp-based community platform.
- Applied systems thinking to solve community management challenges through software solutions.

Full-Stack Software Engineer , Cape Town

FARO Africa | Aug 2024 – Nov 2025

- Extended ERPNext (Python/JS) to improve workflows, pricing logic, and operational reporting (SQL) → reduced manual reporting time.
- Built Expo mobile app with NFC (ISO 14443-4 APDUs) for e-paper price tags → enabled real-time price updates.
- Provisioned AWS infrastructure with Pulumi; deployed Inngest and PayloadCMS services.
- Introduced LLM-assisted development workflows → improved debugging speed and code review throughput.

Embedded Software Engineer , Remote

Ingenics Digital GmbH (through ViVa Outsourcing) | Mar 2023 – May 2024

- Designed event-driven FSM for I2C-based embedded system (C++/FreeRTOS) → maintainable architecture, fewer bugs.
- Developed ESP32 applications (C/C++, ESP-IDF) with configurable BLE stack and FOTA updates.
- Built Python tooling for serial/BLE communication and hardware mocks → accelerated testing workflows.

Software Developer , Cape Town

UMAN Technologies | Mar 2021 – Dec 2022

- Implemented SOME/IP RPC services; optimized performance bottlenecks with perf.
- Built IPC/RPC layers (C++/Python) using Cap'n Proto; led small team using AGILE practices.

Junior Lecturer , Potchefstroom

North-West University | Feb 2020 – Dec 2020

- Lectured Python and C++ programming for first-year IT students (remote and in-person).

Technical Application Scope

Primary	C/C++ Python FreeRTOS ESP32/ESP-IDF NFC APDU (ISO 14443-4)
Secondary	NimBLE UART/SPI/I2C TinyFrame BLE Hardware-in-the-Loop Testing
Foundational	SQL Docker CI/CD Git Linux
Other	React Phoenix AWS Pulumi ERPNext

Education

B.Eng. Computer and Electronic Engineering

North-West University | 2020

Focus on embedded systems, software engineering, and electronic design · NFC payment emulator (Kotlin/Android), STM32 firmware (C), PID motor controller (Arduino) · Data analysis with Python/Pandas using linear regression, correlation and machine learning